

Eliud Koto *AI Research Engineer*

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LinkedIn GitHub Google Scholar

Objective

AI research engineer with a background in statistics and programming. I enjoy using machine learning and statistical methods to understand problems and build practical solutions. I completed an MSc in Mathematical Sciences, specialising in Artificial Intelligence (AI) for Science, at AIMS South Africa as a Google DeepMind Scholar. During the programme, I worked on projects spanning energy efficiency in mobile networks, responsible AI, African language models and AI safety. I am currently working as an AI Research Engineer, where I continue to develop my skills in building and applying machine learning solutions to real-world problems. I am exploring different areas of AI while building deeper expertise in the areas that interest me most.

Education

MSc Mathematical Sciences (AI for Science), *Stellenbosch University* Sep 2025 – July 2026
African Institute of Mathematical Sciences (AIMS) Western Cape, South Africa
Graduated with Distinction

Relevant Coursework: Bayesian Inference, Monte Carlo Methods for Machine Learning, Applied Machine Learning at Scale, Matrix and Tensor Factorization, CUDA Programming for NVIDIA GPUs, Explainable AI (XAI).

BSc (Statistics and Programming), *Kenyatta University* Sep 2015 – Dec 2020
Second Class (Upper Division)

Publications

Koto, E. N., du Toit, J., Stoltz, A., & du Preez, J. (2026). *A Peer-Relative Representation Learning Framework for Energy Inefficiency Identification in Mobile Network Sites.* 

Okunowo, S., Koto, E., & Misker, D. (2026). *Confidently Wrong: Measuring and Mitigating Calibration Risks in LLMs for African Languages.* Presented at Deep Learning Indaba 2026 (workshop and poster). 

Work Experience

Data Scientist, *Onfon Mobile* Sep 2023 – Aug 2025
Nairobi, Kenya

- Developed a credit scoring model using customer transaction and behavioural data to determine the appropriate smartphone financing level for each customer, reducing non-performing loans by 15%.
- Developed statistical and machine learning models using Python and PySpark for fraud detection, helping reduce fraud to below 2%.
- Conducted field research in Lesotho and South Africa to better understand the financial behaviour of customers outside the formal banking system and inform the development of credit and risk models.

Data Analyst, *Infotrace Analytics* Jan 2021 – Aug 2023
Nairobi, Kenya

- Developed Tableau and Power BI dashboards and reports, increasing BI consumption by 13% among users.
- Automated data collection and reporting workflows, reducing manual work and improving access to timely, reliable data.
- Analysed business data to identify trends and provide insights that informed operational decisions.
- Mentored and trained colleagues in data analysis and visualisation, improving team efficiency and confidence in working with data.

Research Projects

Representation Learning for Energy-Efficient Mobile Networks (2026)🔗

- Developed a representation learning framework for detecting energy-inefficient mobile network sites using PyMDE and graph-based embeddings.
- Designed a peer-relative anomaly detection method that compares each site against structurally similar peers rather than global thresholds.
- Generated realistic synthetic mobile network datasets to evaluate robustness under varying anomaly contamination levels.
- Benchmarked the proposed approach against Isolation Forest, Local Outlier Factor, Autoencoder, Random Forest residuals, and Huber regression.

Human-in-the-Loop Student Performance Prediction (2026)🔗

- Developed a student performance prediction framework incorporating uncertainty estimation for trustworthy AI.
- Implemented selective prediction, routing low-confidence predictions to human reviewers.
- Evaluated the trade-off between prediction coverage and model reliability under human oversight.
- Explored practical human-AI collaboration strategies for decision support in educational settings.

Multilingual Polarization Detection (SemEval 2026)🔗

- Participated in the SemEval-2026 shared task on multilingual polarization detection.
- Fine-tuned transformer-based language models for multilingual text classification.
- Performed data preprocessing, model training, and evaluation using standard NLP classification metrics.

Movie Recommendation System (2025)🔗

- Built a collaborative filtering recommendation system using matrix factorisation with Alternating Least Squares (ALS).
- Trained and evaluated recommendation models on large-scale user-item interaction data.
- Assessed recommendation quality using Precision@K, Recall@K, and Mean Average Precision (MAP).
- Learned latent user and item representations for personalised movie recommendations.

Achievements & Awards

- Google DeepMind Scholar, African Institute for Mathematical Sciences (2025–2026).
- Africa Regional Winner, Global South AI Safety Hackathon (2026), organised by Apart Research and Schmidt Sciences.

Skills

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| • Python, PyTorch, PySpark, R | • CUDA Programming (NVIDIA GPUs) |
| • Machine Learning & Deep Learning | • Data Engineering & MLOps |
| • Statistical Modelling & Bayesian Inference | • Data Analysis & Visualisation |
| • Reinforcement Learning | • Scientific Computing |
| • Natural Language Processing (LLMs) | • Research & Technical Writing |
| • Explainable AI (XAI) | • Problem Solving & Critical Thinking |

References

Available upon request.